

Enhancing Embodiment and Spatial Awareness in VR with Hybrid Visual Presentation via Binocular Rivalry

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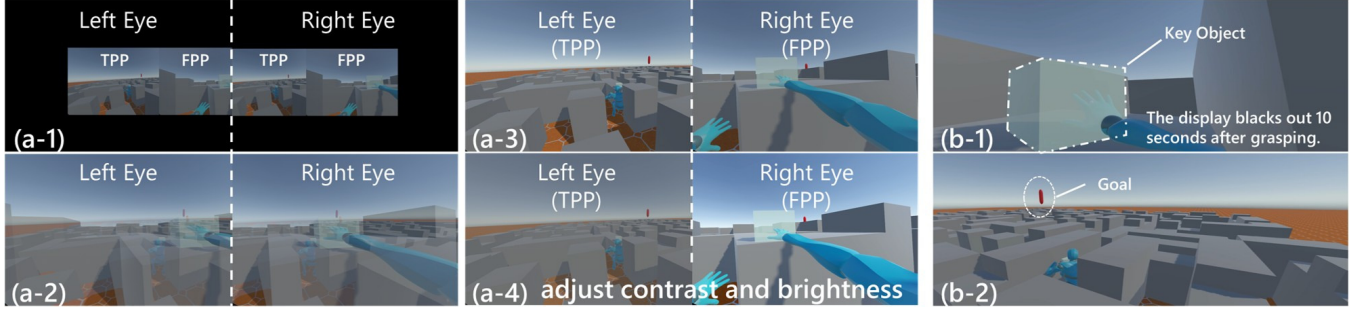


Figure 1: Experimental setup and conditions. (a) Four Visual Presentations: (a-1) Split Presentation, (a-2) Overlay Presentation, (a-3) Binocular Rivalry (BR) Presentation, and (a-4) Binocular Rivalry with Visual Enhancement (BR-V) Presentation. (b) The experimental environment: (b-1) Key object, and (b-2) an example of a maze.

Abstract

This study proposes a novel method leveraging the voluntary perceptual control inherent in binocular rivalry to combine the high sense of embodiment of a first-person perspective and the strong spatial awareness of a third-person perspective. Our method achieves a superior balance of embodied sensation and spatial awareness compared to conventional simultaneous-presentation methods. Furthermore, we show that using visual enhancement during binocular rivalry not only makes a perspective easier to perceive but also positively impacts both embodied sensation and spatial awareness.

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1 Introduction

While First-Person Perspective (FPP) in Virtual Reality (VR) provides a high degree of sense of embodiment, it often lacks comprehensive spatial awareness of the surrounding environment. Conversely, a Third-Person Perspective (TPP) can enhance spatial awareness by allowing users to perceive their avatar's actions in relation

to the environment, but it often diminishes the sense of embodiment. This highlights a fundamental trade-off between sense of embodiment and spatial awareness.

To address this trade-off, we propose a novel visual presentation method that leverages binocular rivalry. We investigate whether this technique can effectively combine both a high sense of embodiment and enhanced spatial awareness within a VR experience, particularly in comparison to previously proposed simultaneous presentation methods[Miura et al. 2021; Schjerlund et al. 2022].

The findings are applicable to multifaceted tasks, such as remote robot operation, that require precise FPP manipulation and simultaneous TPP monitoring. Our method could also be applied to other research areas using multiple viewpoints, such as creating weight illusions via the Proteus effect[Muraishi et al. 2024].

2 Proposed Method

Our proposed method leverages binocular rivalry to balance sense of embodiment and spatial awareness. Binocular rivalry occurs when the brain is presented with sufficiently different images simultaneously to each eye, causing the perceived image to alternate [Charles 1838]. Users can voluntarily control their perception, with higher contrast or brightness images being perceived more easily [Mueller and Blake 1989; Paffen and Alais 2011].

Leveraging these properties, our method presents a different video feed to each eye. This enables users to consciously switch their perception between the two perspectives. We also dynamically adjust contrast and brightness based on the specific task, making the more appropriate perspective easier for the user to perceive. This approach allows for a more natural and seamless switching of views within the user's mind, reflecting their intentions more effectively than conventional methods.

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3 Experiment

We evaluated four visual presentation conditions (Figure 1(a)) that simultaneously present FPP and TPP to combine high sense of embodiment with high spatial awareness.

The Split Presentation (Figure 1(a-1)) divides the screen, presenting each perspective to a different side. The Overlay Presentation (Figure 1(a-2)) composites both perspectives with 0.5 transparency, showing the merged image to both eyes. The BR Presentation (Figure 1(a-3)) induces binocular rivalry by presenting one perspective to each eye. The BR-V Presentation (Figure 1(a-4)) extends BR Presentation by dynamically adjusting the contrast and brightness of each perspective to visually emphasize the more relevant view based on the user's actions.

A total of 11 participants (8 male, 3 female, aged 21-25) took part in the study. We employed a within-subject design, and the presentation order of all conditions was fully randomized to control for potential sequence effects. The task was to navigate a VR maze, grasping a key object and carrying it to the goal (Figure 1(b)). We assumed that the FPP would be most useful for the key-grasping action, while the TPP would be more suitable for navigating the maze. Therefore, in the BR-V Presentation condition, the system automatically determined which perspective to emphasize.

After completing the task, participants answered a 9-item questionnaire on a 7-point Likert scale. The questionnaire, which drew from prior studies and unique questions, included six questions on sense of embodiment (two for each sub-component: body-ownership, agency, and self-location)[Gonzalez-Franco and Peck 2018], two on spatial awareness, and one on viewpoint switching burden. The time taken to complete the task was also recorded.

4 Results

We analyzed the mean response values of the questionnaire using a Friedman test followed by Holm's post-hoc test. Figure 2(a)-(d) presents the results for sense of embodiment and spatial awareness. We present a portion of the notable findings here.

Figure 2(a)-(c) shows the results for sense of embodiment. Our binocular rivalry-based methods (BR and BR-V Presentation) showed a significantly higher sense of embodiment than the Split Presentation. When compared to the Overlay Presentation, our methods were superior in sense of self-location, as shown in Figure 2(c). Furthermore, between the two binocular rivalry-based methods, the BR-V presentation yielded a significantly higher sense of agency than the BR presentation.

Regarding spatial awareness, as shown in Figure 2(d), the BR-V Presentation was significantly higher than the Split Presentation. We also analyzed the normalized task completion time, as shown in Figure 2(e), normalized by the median time for each maze difficulty. BR-V Presentation resulted in significantly shorter completion times than both Split and BR Presentations.

As shown in Figure 2(f), users found it significantly easier to switch perceptions with the BR-V Presentation than with the BR Presentation.

These results lead to several conclusions. First, the use of binocular rivalry-based methods suggests a potential to achieve both a high sense of embodiment and high spatial awareness simultaneously. This is likely because users could actively select between

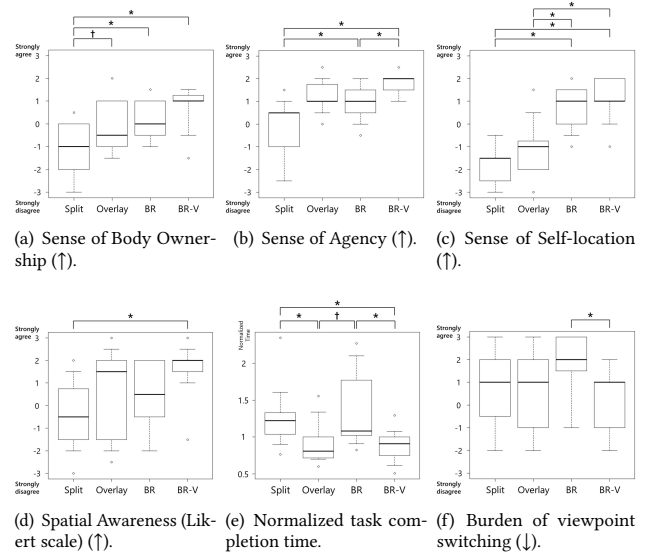


Figure 2: The Results of Sense of Embodiment and Spatial Awareness. (* : $p < 0.05$, † : $p < 0.1$)

the two perspectives based on the task while the benefits of the other perspective were maintained in the background. Second, the effectiveness of visual enhancement is evident. The superior performance of the BR-V Presentation over the BR Presentation indicates that visual enhancement does more than simply make a perspective easier to perceive; it yields a higher overall benefit. By changing the contrast and brightness during binocular rivalry, we speculate that users could perceive their intended perspective more stably. This likely contributed to increased task efficiency, improved spatial awareness, and a higher sense of agency due to a clearer and more stable perception of their actions.

For future work, we plan to apply our method to other tasks, compare it with other techniques, and explore visual enhancements using other binocular rivalry characteristics.

References

- Wheatstone Charles. 1838. XVIII. Contributions to the physiology of vision. -Part the first. On some remarkable, and hitherto unobserved, phenomena of binocular vision. *Philosophical Transactions of the Royal Society of London* 128 (12 1838), 371–394.
- Mar Gonzalez-Franco and Tabitha C. Peck. 2018. Avatar Embodiment. Towards a Standardized Questionnaire. *Frontiers in Robotics and AI* 5 (2018).
- Reiji Miura, Shunichi Kasahara, Michiteru Kitazaki, Adrien Verhulst, Masahiko Inami, and Maki Sugimoto. 2021. MultiSoma: Distributed Embodiment with Synchronized Behavior and Perception. In *Proceedings of the Augmented Humans International Conference 2021*. 1–9.
- T. J. Mueller and R. Blake. 1989. A fresh look at the temporal dynamics of binocular rivalry. *Biol. Cybern.* 61, 3 (July 1989), 223–232.
- Yusei Muraishi, Joichiro Muraoka, Naoki Hashimoto, and Mie Sato. 2024. Examination of effect from a viewpoint position on weight illusion caused by impression change. In *International Workshop on Advanced Imaging Technology (IWAIT) 2024*, Vol. 13164. SPIE, 131642R.
- Chris Paffen and David Alais. 2011. Attentional Modulation of Binocular Rivalry. *Frontiers in Human Neuroscience* 5 (2011).
- Jonas Schjerlund, Kasper Hornbæk, and Joanna Bergström. 2022. OVRlap: Perceiving Multiple Locations Simultaneously to Improve Interaction in VR. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Article 355, 13 pages.